



1.1.4 Wrap-Up: Reflection

1. Rate yourself based on what you understood in today's lesson.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can define parallel lines.	
I can define perpendicular lines.	
I can define a right angle.	

2. Rate yourself based on how you feel about the following statements.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am willing to try new things.	
I persevere when something is challenging.	



Unit 1, Lesson 2: Defined and Undefined Terms – Part 2



Benchmark

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.



Learning Targets

- ☐ I can understand the three undefined terms of geometry.
- ☐ I can use rigid motions to define a straight angle.
- ☐ I can experiment with rigid motions on a plane.



1.2.1 Warm-Up: Would You Rather?

1. Would you rather have pennies to match your weight or quarters to match your height?
2. Explain your reasoning.



1.2.2 Exploration: Definitions

1. Use two sheets of patty paper to complete the following.
 - Draw $\overline{HN}$ on one piece of patty paper.
 - Draw $\overline{BL}$ on the other piece of patty paper.
 - Lay the piece of patty paper with $\overline{HN}$ drawn on it on top of the piece with $\overline{BL}$ drawn on it. Place the papers so that the points H and B are on top of each other and the line segments coincide.
 - Place your writing utensil on endpoint H .
 - Rotate the top sheet of patty paper until the two line segments make an obtuse angle.
2. Draw the obtuse angle you created below.
3. When you rotated the top sheet, did the line segment change in size?
4. When you rotated the top sheet, did the line segment change in position?

5. Which part(s) of the obtuse angle has two points?
6. Rotate the top sheet of patty paper until the two line segments make a straight line.
7. Name the resulting line segment using the endpoints.
8. Work with your partner to estimate the number of degrees you rotated the top sheet of patty paper from its original position on top of line segment BL .



A **straight angle** is an angle measuring exactly 180° .

9. Based on the rotation and the definition of a straight angle, what “fact” about a line can be added to the “Facts about Lines” in the previous lesson?



1.2.3 Bring It Together: Undefined Terms



A **definition** is an exact statement, written formally, of the meaning or description of a word.

There are three terms in geometry that are considered “undefined terms.”

A term is considered undefined if, when explaining the term, the explanation relies on the term itself or a close synonym. This is sometimes called a circular definition.

1. In the previous lesson you wrote a definition for the term “line.” Review your definition and determine if your definition is considered circular.

Although we have “facts about lines,” “line” is an undefined term in geometry. The other two undefined terms in geometry are “point” and “plane.” Look back through the previous lesson to notice how often these terms were used without providing a definition.

2. Work with your partner to label each description as a line, plane, or point.

Description	Undefined Term
has two dimensions	☐ Line ☐ Point ☐ Plane
has no length, no width, and no height	☐ Line ☐ Point ☐ Plane
is often drawn as a four-sided figure	☐ Line ☐ Point ☐ Plane
has infinite length, no width, no height	☐ Line ☐ Point ☐ Plane
named by an ordered pair in the coordinate plane	☐ Line ☐ Point ☐ Plane
indicates a location or position	☐ Line ☐ Point ☐ Plane
has one dimension	☐ Line ☐ Point ☐ Plane
has infinite length, infinite width, and zero height	☐ Line ☐ Point ☐ Plane
has no dimension or actual size	☐ Line ☐ Point ☐ Plane



1.2.4 Wrap-Up: Reflection

1. Rate yourself based on what you understood in today's lesson.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can understand the three undefined terms of geometry.	
I can define a straight angle.	

2. Rate yourself based on how you feel about the following statements.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am a student learner.	
I am a mathematician.	